Final Project: Healthy Minds

S&DS 5360 | Multivariate Statistics

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Introduction

Over the course of the semester, we studied the Healthy Minds Network's latest 2024–2025 Healthy Minds Study (HMS) student survey. The HMS is a population-level, web-based survey administered by the HMN to assess mental health, service utilization, and related factors among post-secondary students. Since its inception over 15 years ago, the study has collected more than $935,000$ responses from over 675 institution nationwide. The proposed research aims to examine the multivariate relationship between sexual and gender minority (GSM) identities, digital environment exposure, and mental health outcomes in the 2024-25 college student population. Utilizing the Demographics and Mental Health Status standard modules, this study will investigate how latent factors of psychological distress—comprising the PHQ-9 (Depression), GAD-7 (Anxiety), and UCLA Loneliness Scale—vary across intersectional identities.

Planned Analysis (Edit)

To understand the latent structures and relationships within this data, we will perform four multivariate techniques:

Principal Component Analysis (PCA): To reduce our correlated mental health variables into a single index of psychological distress.

Ordination (CA & NMDS): To non-linearly map the topological structure of specific, individual mental health symptoms and evaluate how they correlate with environmental stressors.

MANOVA: To evaluate how specific demographics (sexual orientation and therapy utilization) impact these mental health outcomes simultaneously.

Discriminant Analysis: To determine if these psychological profiles can accurately predict a student's gender and sexual identity group.

Variables

The resulting dataset features $n = 84,735$ individual respondents. For the purpose of this analysis, we will focus on questions that fall under the Standard Modules. Specifically, we will use composite continuous variables, individual symptom frequencies, and categorical predictors:

- Composite Scores: flourish (8-56), deprawsc (0-27), anx_score (0-21), lonesc (3-9), and ed_sde (0-5).
  - flourish: (Continuous) 8 statements using a 1-7 scale to indicate agreement with items like "I lead a purposeful and meaningful life." (min = 8, max = 56) .
  - dprawsc: (Continuous) 9 questions over the last 2 weeks regarding problems like "Feeling down, depressed or hopeless." (min = 0, max = 27) .
  - anx_score: (Continuous) 7 questions over the last 2 weeks regarding problems like "Feeling nervous, anxious or on edge." (min = 0, max = 21) .
  - lonesc: (Continuous) 3 questions regarding lacking companionship and feeling isolated. (min = 3, max = 9) .
  - ed_sde: (Continuous) 5 questions regarding disordered eating desires and behaviors. (min = 0, max = 5) .
- Individual Symptoms (Ordinal 1-4): Depression (phq9_1 to phq9_9) and Anxiety (gad7_1 to gad7_7).
- Environmental/Lifestyle: exerc (exercise hours), internet_1 (time online), fincur (financial stress), and sleep_wknight (hours of sleep).
- Identity Groups / Categorical Predictors: queer (Binary 0/1 for sexual orientation), ther_any (Binary 0/1 for therapy utilization), and identity_group (Categorical: Cis-Het, Cis-LGBQ+, TGNC).

```r
# Load data
hms_data <- read_sav("../data/hms_data.sav", encoding = 'latin1')

symptoms <- c(paste0('phq9_', 1:9), paste0('gad7_', 1:7))
environment <- c('age', 'lonesc', 'exerc', 'internet_1', 'fincur')

# Unified Data Cleaning (NAs are preserved globally)
df <- hms_data %>%
  mutate(across(c(gender_trans, gender_queer, gender_nonbin, gender_selfID,
                  sexual_h, sexual_l, sexual_g, sexual_bi, sexual_queer,
                  sexual_quest, sexual_pan, sexual_asexual),
                ~replace_na(., 0))) %>%
```

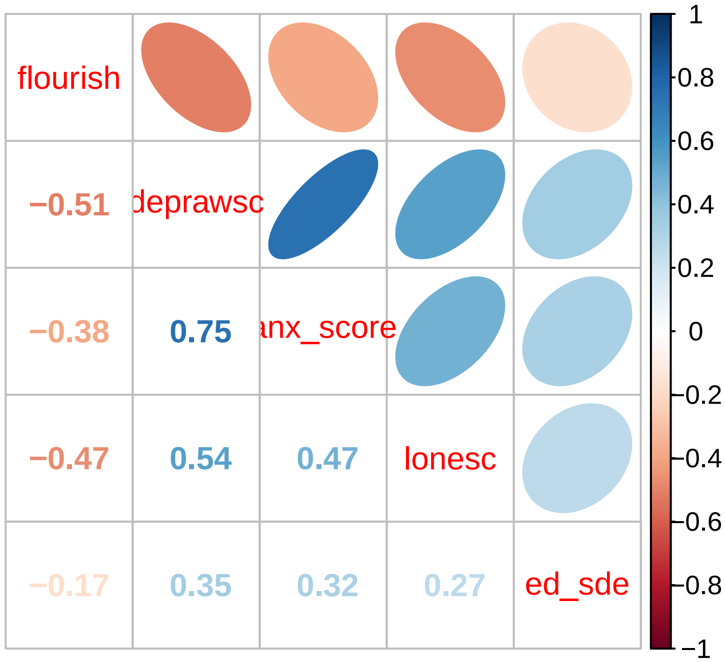
```r
  mutate(
      trans = (gender_trans == 1 | gender_queer == 1 | gender_nonbin == 1 |
               gender_selfID == 1),
      queer_flag = (sexual_l == 1 | sexual_g == 1 | sexual_bi == 1 |
                    sexual_queer == 1 | sexual_quest == 1 | sexual_pan == 1 |
                    sexual_asexual == 1),
      identity_group = case_when(
          trans ~ "TGNC",
          !trans & queer_flag ~ "Cis-LGBQ+",
          !trans & !queer_flag & sexual_h == 1 ~ "Cis-Het",
          TRUE ~ NA_character_
      )
  ) %>%
  mutate(
      identity_group = as.factor(identity_group),
      ther_any = as.factor(ther_any),
      queer = as.factor(queer_flag),
      sleep_wknight = as.numeric(sleep_wknight),
      deprawsc = as.numeric(deprawsc),
      flourish = as.numeric(flourish)
  )
```

Descriptive Plots and Transformations

Our initial correlation plot suggests strong linear relationships between our composite variables. There appears to be a distinct cluster of positive correlations between depression (deprawsc), anxiety (anx_score), and loneliness (lonesc), which are all strongly negatively correlated with flourishing (flourish).

```r
pca_data <- df %>%
  dplyr::select(flourish, deprawsc, anx_score, lonesc, ed_sde) %>%
  drop_na()

corr_matrix <- cor(pca_data)
corrplot.mixed(corr_matrix, lower = "number", upper = "ellipse")
```



We sample $5,000$ observations for our matrix plot, as plotting all $68,062$ observations would result in overplotting and be computationally intensive. We see strong structure and skewness in the Q-Q plot, suggesting that the data are not multivariate normal. At the cost of interpretability, we transform the variables, not only to hopefully attain multivariate normality (which is not required for PCA), but to improve linearity and reduce outlier influence:

- Square Root: deprawsc and anx_score to correct for right-skew.
- Power Transformation (x^2): flourish to correct for left-skew.

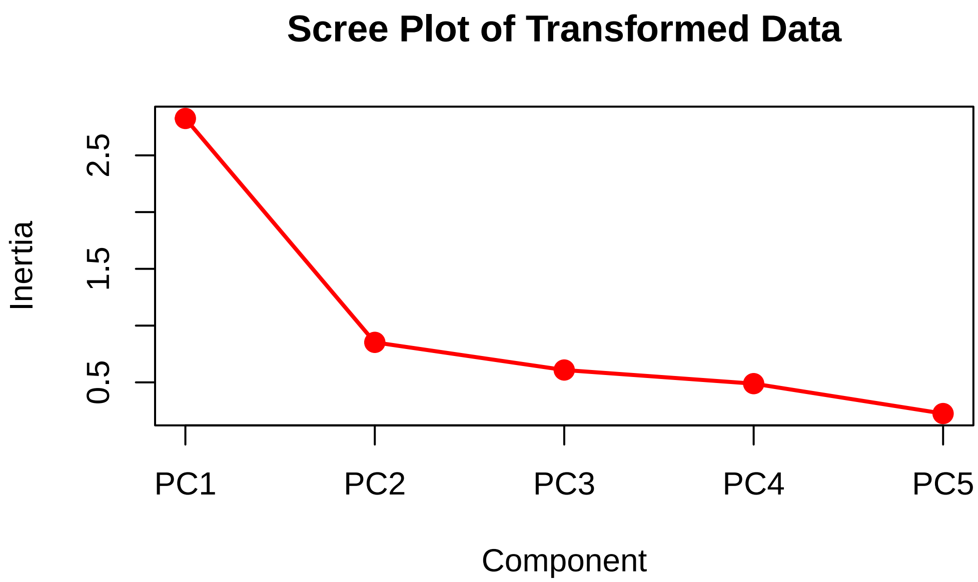
```
df_transform <- df %>%
    mutate(
        deprawsc_sqrt = sqrt(deprawsc + 0.5),
        anx_sqrt = sqrt(anx_score + 0.5),
        flourish_sq = flourish^2
    )
```

Method 1: Principal Component Analysis

Based on these correlations, combined with the aforementioned ratio of $n = 68,062$ observations against $p = 5$ variables (a response-to-variable ratio of over $13,600 : 1$), PCA is expected to perform well.

```r
pca_data_transform <- df_transform %>%
  dplyr::select(flourish_sq, deprawsc_sqrt, anx_sqrt, lonesc, ed_sde) %>%
  drop_na()

pca <- prcomp(pca_data_transform, scale. = TRUE)
screeplot(pca, type = "lines", col = "red", lwd = 2, pch = 19, cex = 1.2,
          main = "Scree Plot of Transformed Data")
```



Note that the first principal component (PC1) captures 56% of the variance. Adding PC2 increments this to 74% of the variance, and PC3 brings us to 86%. However, note that only one eigenvalue is greater than one: $\lambda_{\text{PC1}} = 2.82$. There is a clear elbow after the first principal component, which suggests keeping only PC1, just as suggested by the eigenvalue > 1 rule.

```r
round(pca$rotation[, 1:2], 2)
```

```
                PC1   PC2
flourish_sq    0.42 -0.43
deprawsc_sqrt -0.53  0.01
anx_sqrt      -0.49 -0.06
lonesc        -0.45  0.24
ed_sde        -0.31 -0.87
```

Based on the loadings for PC1, we see two predominant axes emerge: a measure of general well-being and a measure of psychological distress. Observe that flourishing is the only variable with a positive loading (0.42), indicating that a higher score on the flourishing scale contributes directly to a higher PC1 score. Conversely, the remaining variables have negative loadings, with depression having the largest effect (-0.53) on a reduced PC1 score. Effectively, we can interpret PC1 to be a measure of a student's psychological health.

Method 2: Ordination

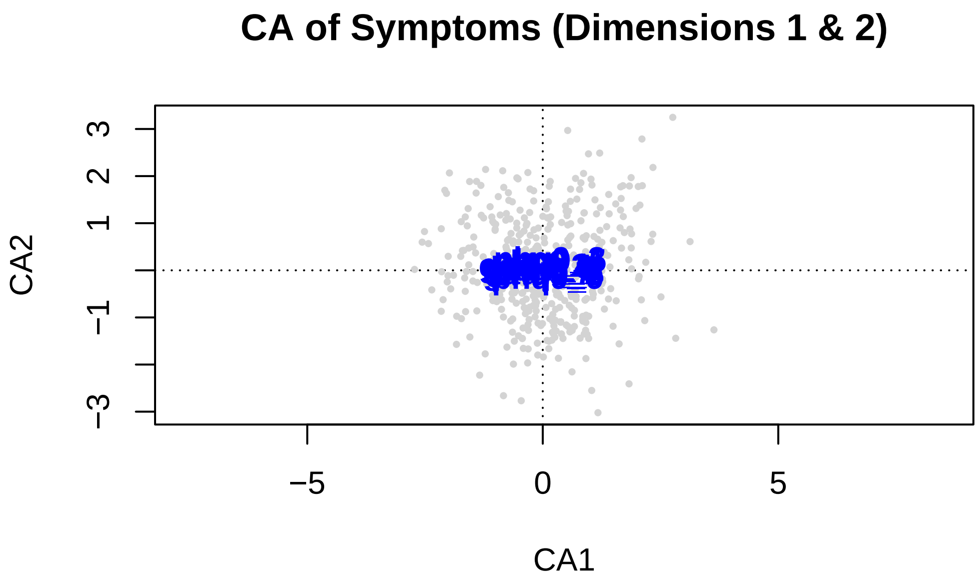
While PCA gave us a broad, linear index of psychological health using composite scores, we next use Ordination to map the specific, topological relationships between individual symptoms. We begin with a Correspondence Analysis (CA).

```
set.seed(4747)
df_subset <- df %>%
  dplyr::select(all_of(symptoms), all_of(environment)) %>%
  drop_na() %>%
  sample_n(500)

hms_symptoms_subset <- df_subset %>% dplyr::select(all_of(symptoms))
hms_env_subset <- df_subset %>% dplyr::select(all_of(environment))

hms_ca <- cca(hms_symptoms_subset)

plot(hms_ca, choices = c(1, 2), type = "n",
     main = "CA of Symptoms (Dimensions 1 & 2)")
points(hms_ca, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_ca, display = "species", col = "blue", cex = 1, font = 2)
```



We see that our total inertia is 0.1073. Upon visualizing the first two dimensions, there is no clear evidence of an arch effect (data snaking). The lack of snaking and the tight clustering near the origin indicate that mental health symptoms measured by PHQ-9 and GAD-7 are highly comorbid; students experience these symptoms simultaneously rather than turning over from one distinct symptom to another.

Because our symptom scales rely on ordinal frequency with high rates of joint absences, we pivot to Non-Metric Multidimensional Scaling (NMDS) using a Bray-Curtis distance metric to better preserve the rank-order topology.

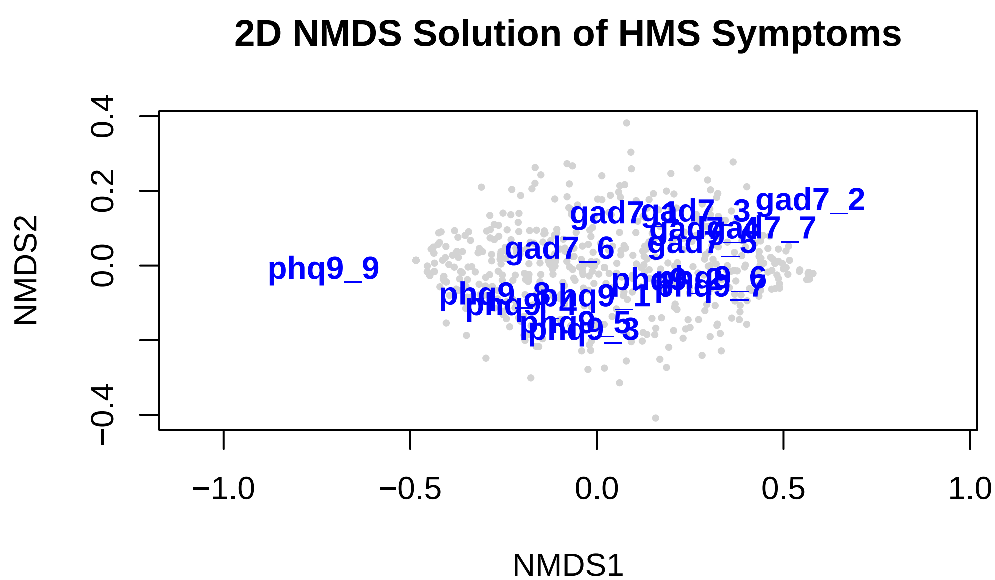
```r
hms_nmds2 <- metaMDS(hms_symptoms_subset, distance = "bray", k = 2, trymax = 50)
```

```
Run 0 stress 0.1346835
Run 1 stress 0.1489844
Run 2 stress 0.1487696
Run 3 stress 0.1491718
Run 4 stress 0.1518426
Run 5 stress 0.1497452
Run 6 stress 0.1486803
Run 7 stress 0.1480797
Run 8 stress 0.1479384
Run 9 stress 0.1481843
Run 10 stress 0.1428595
```

```
Run 11 stress 0.1510179
Run 12 stress 0.1491022
Run 13 stress 0.1466674
Run 14 stress 0.148218
Run 15 stress 0.1429756
Run 16 stress 0.1482841
Run 17 stress 0.1497466
Run 18 stress 0.1486081
Run 19 stress 0.1511065
Run 20 stress 0.1459855
Run 21 stress 0.1448897
Run 22 stress 0.1487585
Run 23 stress 0.1477023
Run 24 stress 0.1481875
Run 25 stress 0.1478835
Run 26 stress 0.1493231
Run 27 stress 0.1467557
Run 28 stress 0.1440557
Run 29 stress 0.1501141
Run 30 stress 0.1419421
Run 31 stress 0.1483056
Run 32 stress 0.1474792
Run 33 stress 0.1449226
Run 34 stress 0.1479583
Run 35 stress 0.1484182
Run 36 stress 0.1455798
Run 37 stress 0.1497667
Run 38 stress 0.144032
Run 39 stress 0.1478023
Run 40 stress 0.1505274
Run 41 stress 0.1491837
Run 42 stress 0.1505833
Run 43 stress 0.1437823
Run 44 stress 0.1509596
Run 45 stress 0.1468367
Run 46 stress 0.1409706
Run 47 stress 0.1487545
Run 48 stress 0.1459431
Run 49 stress 0.1454545
Run 50 stress 0.1484012
*** Best solution was not repeated -- monoMDS stopping criteria:
    16: no. of iterations >= maxit
    26: stress ratio > sratmax
```

```
8: scale factor of the gradient < sfgrmin
```

```
plot(hms_nmds2, type = "n", main = "2D NMDS Solution of HMS Symptoms")
points(hms_nmds2, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_nmds2, display = "species", col = "blue", cex = 1, font = 2)
```

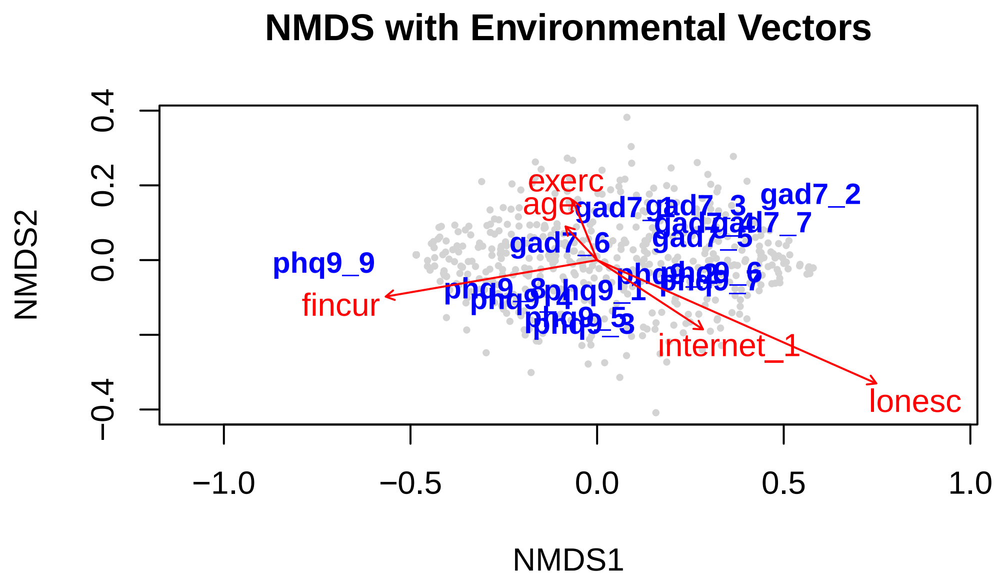


The 2-dimensional plot (Stress = 0.1347, well below the 0.20 threshold) reveals a clear mapping of the symptom space. Observe the extreme observation phq9_9 on the far left-hand side, which specifically measures suicidal ideation and self-harm. A less drastic, but still notable, distinction is the vertical axis, which separates physical symptoms like phq9_3 (sleep) and phq9_4 (fatigue) at a lower elevation from GAD-7 anxiety variables higher up.

We can overlay our environmental continuous variables onto this 2D NMDS solution to see what drives this symptom topology.

```
fit_env <- envfit(hms_nmds2, hms_env_subset, permutations = 999)

plot(hms_nmds2, type = "n", main = "NMDS with Environmental Vectors")
points(hms_nmds2, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_nmds2, display = "species", col = "blue", cex = 0.9, font = 2)
plot(fit_env, col = "red", lwd = 2)
```



Looking at the direction of each arrow, lonesc (loneliness, $R^2 = 0.3228$) and internet_1 point down and to the right, which is where a blend of general anxiety and depression symptoms lie. Meanwhile, fincur (financial stress) pulls hard to the left, acting along the primary axis where the severe outlier phq9_9 sits, indicating it is a major driver behind more severe ideation.

Method 3: Two-Way MANOVA

With the topology of student mental health and environmental drivers mapped, we investigate how demographic and lifestyle factors independently and interactively influence these outcomes. We run a two-way Type III MANOVA to examine the effect of therapy and sexual orientation on depression and flourishing.

```
df_manova <- df %>%
  dplyr::select(deprawsc, flourish, ther_any, queer, sleep_wknight) %>%
  drop_na()

manova_fit <- lm(cbind(deprawsc, flourish) ~ ther_any * queer, data = df_manova)
Manova(manova_fit, type = 'III')
```

```
Type III MANOVA Tests: Pillai test statistic
```

```
                Df test stat approx F num Df den Df    Pr(>F)    
(Intercept)      1    0.95461   328294      2  31221 < 2.2e-16 ***
ther_any         1    0.01820      289      2  31221 < 2.2e-16 ***
queer            1    0.03175      512      2  31221 < 2.2e-16 ***
ther_any:queer  1    0.00111       17      2  31221 2.954e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Using Pillai's Trace, all efforts were statistically significant ($p < 0.001$). The F statistics reveal that the main effects of sexual orientation ($F = 513$) and therapy ($F = 250$) are substantially larger than their interaction ($F = 10$).

We add average hours of sleep per weeknight (sleep_wknight) as a continuous predictor.

```r
model_cov <- lm(cbind(deprawsc, flourish) ~
                ther_any + queer + sleep_wknight, data = df_manova)

Manova(model_cov, type = "III")
```

```
Type III MANOVA Tests: Pillai test statistic
              Df test stat approx F num Df den Df    Pr(>F)    
(Intercept)    1   0.66822  31440.6      2  31221 < 2.2e-16 ***
ther_any       1   0.02179    347.8      2  31221 < 2.2e-16 ***
queer          1   0.04458    728.4      2  31221 < 2.2e-16 ***
sleep_wknight  1   0.08140   1383.3      2  31221 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Somewhat surprisingly, sleep is not only a significant predictor in our multivariate analysis: it is also the largest ($F = 1{,}348$). We see that for every additional hour of sleep a student gets, their depression score drops by 1.19 points, and their flourish score increases by 0.98 points.

Method 4: Discriminant Analysis

Given the significant multivariate differences between demographic groups established in MANOVA, we applied Discriminant Analysis to see if mental health scores alone can predict identity group membership (Cis-Heterosexual, Cis-LGBQ+, and TGNC).

```r
df_lda <- df %>%
  dplyr::select(identity_group, deprawsc, anx_score, lonesc, flourish) %>%
  drop_na()

# Run Cross-Validated LDA
lda_cv <- lda(
    identity_group ~ deprawsc + anx_score + lonesc + flourish,
    data = df_lda,
    prior = c(1/3, 1/3, 1/3),
    CV = TRUE)

lda_cv_acc <- round(sum(diag(prop.table(table(df_lda$identity_group, lda_cv$class)))), 4)
cat("LDA CV Accuracy:", lda_cv_acc, "\n")
```

```
LDA CV Accuracy: 0.5167
```

```r
# Confusion Matrix
conf_matrix <- table(Actual = df_lda$identity_group, Predicted = lda_cv$class)
conf_matrix
```

```
           Predicted
Actual       Cis-Het Cis-LGBQ+  TGNC
  Cis-Het      28363      6331 10097
  Cis-LGBQ+     7040      3106  6241
  TGNC          1498       852  2805
```

Our overall cross-validated accuracy is approximately 52%. The model is most effective at predicting Cis-Het and TGNC, with recalls of 63.3% and 54.4%, accordingly. However, the Cis-LGBQ+ group has a low recall of 19.0%. Most of these students are misclassified as TGNC, illustrating that their psychological health data doesn't form a unique, linearly separable cluster.

Conclusion and Discussion

Through a comprehensive multivariate approach, we established that student mental health is largely driven by a single latent construct of general distress and well-being (PCA). However, digging deeper into the topology of individual symptoms via Ordination (NMDS) revealed a nuanced structure: while symptoms are highly comorbid, severe clinical manifestations like self-harm map differently than general anxiety, and these are distinctly driven by environmental stressors like financial burden and loneliness. Our MANOVA further contextualized this by revealing that sexual minority status and therapy utilization are strong main effects driving distress, though healthy lifestyle factors like sleep can meaningfully offset it. Finally, Discriminant Analysis confirmed that while mental health profiles differ significantly by identity, the lived experiences of Cis-LGBQ+ and TGNC students share enough psychological overlap that they are difficult to linearly separate, underscoring the shared systemic stressors experienced by the broader LGBTQIA+ umbrella.